

# SCHOOL OF COMPUTING (SEM II – 2021/2022)

## SECI2143 – PSDA

### Exercise (Chapter 5 – Part2)

#### QUESTION 1

- (a) During flu outbreak, parents of 2000 babies were surveyed by researchers of a pharmaceutical company to determine if the company's new flu medicine was effective after two days. Among 120 babies who had the flu and were given the medicine, 29 were recovered within two days. Among 280 babies who had the flu but were not given the medicine, 56 recovered within two days. Is there any significant indication that supports the company's claim of the effectiveness of the flu medicine? Use a 0.01 level of significance.

#### QUESTION 2

Past experience indicates that the time required for students in a university to complete a standardize test is a normal random variable with a mean of 35 minutes. If a random sample of 20 students took an average of 33.1 minutes to complete the test with a standard deviation of 4.3 minutes, test the hypothesis, that the average time taken is actually less than 35 minutes.

- (i) State your hypothesis null,  $H_0$  and hypothesis alternative,  $H_1$ . (1 mark)  
(ii) Test the hypothesis given in (i) at the 0.05 significance level. Do you reject or fail to reject the  $H_0$ ? State your conclusion. (5 marks)

#### QUESTION 3

- (a) A random sample of 6 steel beams has a mean compressive strength of 58392 psi (pound per square inch) with its population standard deviation is 648 psi. Use this information and the level of significance  $\alpha = 0.05$  to test whether the true average compressive strength of steel from which this sample came is 58000 psi. (6 marks)
- b) Past data indicate that the variance of measurements made on sheet metal stampings by experienced quality control inspectors is 0.18 (inch)<sup>2</sup>. Such measurements made by an inexperienced inspector could have too large a variance (perhaps because of inability to read instruments properly) or too small a variance (perhaps because unusually high or low measurements are discarded). If a new inspector measures 101 stampings with variance of 0.13 (inch)<sup>2</sup>, test at the 0.05 level of significance whether the inspector is making satisfactory measurements. Assume data is normally distributed. (6 marks)

#### QUESTION 4

- (a) In a research report, Richard H. Weindruch of the UCLA Medical School claims that mice with an average life span of 32 months will live to be about 40 months old when 40% of the calories in their diet are replaced by vitamins and protien. Is there any reason to believe that  $\mu < 40$  if 64 mice that are placed on this diet have an average life of 38 months with a standard deviation of 5.8 months? Use a 0.05 level of significance. (5 marks)
- (b) Test the hypothesis that the average content of container of a particular lubricant is 10 liters if the contents of a random sample of 10 containers are 10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, and 9.8 liters. Use a 0.01 level of significance and assume that the distribution of contents is normal. (10 marks)

#### QUESTION 5

- a) In a research report, Richard H. Weindruch of the UCLA Medical School claims that mice with an average life span of 32 months will live to be about 40 months old when 40% of the calories in their diet are replaced by vitamins and protien. Is there any reason to believe that  $\mu < 40$  if 64 mice that are placed on this diet have an average life of 38 months with a standard deviation of 5.8 months? Use a 0.05 level of significance. (5 marks)
- b) Test the hypothesis that the average content of container of a particular lubricant is 10 liters if the contents of a random sample of 10 containers are 10.2, 9.7, 10.1, 10.3, 10.1, 9.8, 9.9, 10.4, 10.3, and 9.8 liters. Use a 0.01 level of significance and assume that the distribution of contents is normal. (10 marks)